

The CP AI RAG Service — on one page

Grounded, page-cited answers over court documents, as a service.

2,086

documents ingested in the 76-day production pilot; 15,200 queries answered, zero rate-limit errors

5½p

billed LLM cost per answered query in the pilot — 3.6p generation + 1.9p scoring (GPT-4o)

100%

of answers pass a citation guard before delivery; ~96% were groundedness-scored in the pilot

£590K

projected annual run cost at national volume on the current model, pay-as-you-go

Scope

Common Platform systems upload court documents and ask natural-language questions of them. Answers come back with **every claim cited to a document and a page**, plus the evidence passages for verification. A headless capability — no UI of its own — consumed as five HTTP operations behind API Management, contract-first.

Decision support, never decision automation: single-turn Q&A, scoped by caller metadata (e.g. case ID), with document versioning built in. Out of scope today: conversation memory, streaming, non-PDF formats, hard delete.

Responsible AI use

- **Grounding enforced, not asserted.** Answers may only restate the retrieved documents; an answer whose citations fail is flagged with a recorded reason or rejected — never returned silently.
- **Refusal is protected.** "The documents do not contain this" always passes the guard; the model is never pushed to invent.
- **Measured continuously.** Answers are judged for groundedness in production, and an offline harness gates prompt and model changes with evidence — it has already surfaced a conviction-recall risk in the cheaper migration candidate (missed convictions in 4 of 50 evaluated cells).
- **Human verification by design.** Evidence passages travel with every answer, so a person can check each claim against its source.
- **Content filtering is off by deliberate, documented policy** — the corpus is sensitive case material a filter would refuse to read; grounding and scoring compensate.

Technical approach

A fully serverless Azure pipeline, queue-decoupled at every expensive step. Ingestion: Document Intelligence extraction → per-page chunking → embeddings → Azure AI Search hybrid index. Retrieval over-fetches ~50 candidates, then dedup and diversity stages select ~15 evidence chunks for the model (Azure OpenAI GPT-4o today; migration under evaluation).

The model must emit a structured **fact map**; a **citation guard** flags or rejects degraded answers; a judge model scores groundedness to Azure Monitor. Idempotency fencing makes queue retries double-spend-proof; storage access is managed identity only.

Constraints to scaling nationally

- × **Shared-capacity ceilings, no SLA.** The projected national peak draws ~76% of the pay-as-you-go token ceiling, and scoring's peak breaches its own — beyond them, requests fail hard. Hedges: a quota uplift; a 25–50-PTU fast lane (£66K–£132K/yr); self-hosted scoring (~£70K/yr saving).
- × **No priority or fairness yet.** One queue, one model tier: urgent queries can sit behind routine bulk, and one busy consumer can starve others. The priority-first design (DD-42909) is agreed but not yet built.
- × **Model migration is open.** GPT-4o is prior-generation; gpt-5.1 is ~30% cheaper but carries the recall risk and offers no UK-resident processing; Claude buys the recall back at ~2× per-request cost, UK-resident only via AWS Bedrock.
- × **Multi-client isolation is flag-gated** pending cut-over; onboarding many consumers also needs the fairness work above.
- × **Operational residue:** fixed chunking, PDF-first, soft delete only, integration suite not yet in CI.

Cost vs benefit

ANNUAL COST AT NATIONAL VOLUME (2,000 DOCS/DAY)	TOTAL
Fixed platform, common to every option — ingestion, vector DB, functions	£314K
Today's model, PAYG (GPT-4o) — zero infrastructure, no SLA	~£590K
Recommended hybrid — PTU fast lane + PAYG, self-hosted scoring	~£550–620K
Full 24/7 reserved capacity (PTU) — full SLA, over-provisioned here	~£741–790K
Migration: gpt-5.1 path — ~30% below GPT-4o like-for-like ² ; recall risk	~£510K
Migration: Claude path (Sonnet 4.6 + Haiku) — strongest recall; ~£488K with batching	~£661K

GPT-4o rows: billed basis, 14K queries/day. Migration rows: list rate, 20K/day. ² The like-for-like figure bridges the two bases.

The benefit side. Case preparation means finding the passage that matters in bundles running to hundreds of pages — an hour spent locating material is an hour not spent considering it. At national volume the service answers **~5–7M queries a year at ~2–4p of model cost each** (8–12p all-in), each a page-cited, scored answer returned in seconds.

The pilot proved the mechanics: 76 days, £7,219 (~£95/day), unit costs measured from production billing (27.6p/document, 3.6p/answer, 1.9p/score). The projections are arithmetic on measured units, not estimates.

READ THE COSTS HONESTLY

Figures are projections on pilot-measured token shapes; production billing ran ~×1.2 over list. The migration decision is not purely financial: the cheapest path carries a measured recall risk, and UK-residency requirements could force the platform choice.